Comprehensive Manual for Vertebral Labeling and the Identification of Lumbosacral Transitional Vertebrae (LSTV) in Medical Imaging Datasets

1. Introduction: The Clinical and Computational Imperative of Accurate Labeling

The accurate identification and segmentation of vertebral bodies represent a foundational challenge in medical image analysis, bridging the disciplines of clinical radiology, spinal surgery, and computer vision. While the vertebral column typically adheres to a rigid formula—seven cervical, twelve thoracic, and five lumbar vertebrae—anatomical variations occur with significant frequency. Among these, Lumbosacral Transitional Vertebrae (LSTV) are the most prevalent and clinically confounding, affecting between 4% and 30% of the general population.[1] For medical students and annotators tasked with generating "ground truth" datasets for artificial intelligence (AI) training, LSTV constitutes the single largest source of labeling error.

The stakes of this annotation process extend far beyond academic categorization. In the clinical realm, the misidentification of a transitional segment is a primary cause of "wrong-level" spinal surgery, a catastrophic event where an intervention is performed at an incorrect anatomical level (e.g., L4-L5 instead of L5-S1) due to numbering ambiguity.[3] In the computational realm, the performance of object detection models (such as YOLOv8) and segmentation networks (such as U-Net) is directly correlated with the consistency of the training data. If a dataset contains conflicting labels—where one annotator designates a segment as "L5" based on counting from the skull base, while another designates it as "S1" based on sacral morphology—the resulting model will fail to converge or produce low-confidence predictions.[5]

This manual provides an exhaustive, expert-level guide for the identification and labeling of LSTV. It moves beyond simple vertebral counting to establish a multi-planar, multi-landmark protocol that integrates radiologic gold standards with the technical requirements of modern annotation platforms like Roboflow. By adhering to the methodologies outlined herein, annotators will generate datasets that are anatomically rigorous, clinically valid, and computationally robust.

1.1 The Definition of the Problem Space

LSTV encompasses a spectrum of congenital anomalies at the lumbosacral junction, involving either the **sacralization of the L5 vertebra** (where L5 assumes sacral characteristics) or the **lumbarization of the S1 vertebra** (where S1 assumes lumbar characteristics). These anomalies disrupt the standard numbering of the spine, creating a "transitional" segment that defies binary classification. The ambiguity is compounded in isolated lumbar spine imaging (MRI or CT), where the cervical and thoracic reference points are absent, forcing the annotator to rely on probabilistic anatomical landmarks.[1]

This manual serves two distinct but interrelated purposes:

1. **Anatomical Verification:** To provide a decision tree for determining whether a vertebra is L5, S1, or a transitional variant using landmarks like the iliolumbar ligament (ILL), costal facets, and paraspinal vasculature.
2. **Technical Execution:** To translate these anatomical findings into precise annotations (bounding boxes, polygons, keypoints) within the Roboflow interface, ensuring high-quality input for machine learning models.

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2. Anatomical Fundamentals of the Lumbosacral Junction

To annotate the spine effectively, one must first understand the "normal" anatomy and how LSTV deviates from it. The lumbosacral junction is a region of significant biomechanical stress and morphological transformation, where the mobile, lordotic lumbar spine meets the rigid, kyphotic sacrum.

2.1 Standard Vertebral Architecture

In a standard spinal column, the L5 vertebra is the largest of the mobile segments. It possesses a distinct morphology tailored for weight bearing and attachment:

- **Vertebral Body:** Reniform (kidney-shaped) in the axial plane; wedge-shaped (taller anteriorly) in the sagittal plane to support lumbar lordosis.
- **Transverse Processes:** Short, thick, and pyramidal, originating from the junction of the pedicle and lamina. They serve as attachment points for the iliolumbar ligaments but do not articulate with the sacrum.[7]
- **Disc Space:** The L5-S1 disc is typically wedge-shaped and well-preserved in height, facilitating flexion and extension.

2.2 The Spectrum of Transition (LSTV)

LSTV represents a failure of segmentation (in the case of sacralization) or a failure of fusion (in the case of lumbarization) during embryonic development. This results in a vertebra that

shares characteristics of both the lumbar and sacral spine.

2.2.1 Sacralization of L5

In this scenario, the fifth lumbar vertebra fuses to varying degrees with the sacrum.

- **Anatomical Impact:** The transverse processes of L5 become dysplastic and enlarged (mega-apophysis). They may form a pseudo-joint (diarthrosis) with the sacral ala or fuse completely (osseous bridge).
- **numbering Implication:** The patient functionally has only four mobile lumbar vertebrae. However, the transitional segment is embryologically L5. In labeling, it is often designated as "L5 (Transitional)" or "L5 (Sacralized)" depending on the dataset ontology.[1]

2.2.2 Lumbarization of S1

In this scenario, the first sacral segment separates from the sacral mass.

- **Anatomical Impact:** A fully formed intervertebral disc appears between S1 and S2. The S1 vertebral body loses its wedge shape and becomes "squared" (rectangular) in the sagittal plane.
- **Numbering Implication:** The patient functionally has six mobile lumbar vertebrae. The transitional segment is embryologically S1 but functionally L6. Labeling protocols may require this to be tagged as "L6" or "Transitional S1".[1]

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3. The Castellvi Classification: The Coronal Standard

The Castellvi classification system, proposed in 1984, remains the radiological gold standard for describing the morphology of the transverse processes in LSTV. While it was originally developed for plain radiographs, it is best assessed by annotators on **Coronal** MRI sequences or Coronal reformatted CT images. Understanding this classification is critical for distinguishing between "normal" L5 variants and true transitional vertebrae.[8]

3.1 Detailed Breakdown of Castellvi Types

The classification focuses exclusively on the relationship between the transverse process of the potential transitional segment and the sacral ala.

Castellvi Type	Classification	Morphological Features	Biomechanical & Clinical Implications
Type I	**Dysplastic Transverse Process**	Transverse process height (vertical dimension)	**Clinical:** Generally considered asymptomatic. It

		exceeds **19 mm**. There is no articulation or fusion with the sacrum.	does not alter the biomechanics of the lumbosacral junction significantly. **Labeling:** Often labeled as a normal L5, potentially with a Type I attribute tag.
Type II	**Incomplete Transition (Pseudoarticulation)**	The enlarged transverse process forms a diarthrodial joint (pseudo-joint) with the sacrum. This is not a solid fusion; a gap exists.	**Clinical:** Strongly associated with **Bertolotti's Syndrome**. The abnormal joint alters biomechanics, limiting motion and causing stress. Edema may be visible here. **Labeling:** Must be distinguished from Type III. Look for the "joint line."
Type III	**Complete Fusion**	True osseous continuity (bony bridge) between the transverse process and the sacrum. No joint space is visible.	**Clinical:** Provides stability. The motion segment is effectively eliminated, transferring stress to the level above (L4-L5). **Labeling:** Verify on bone-window CT or T1 MRI for continuous marrow

			signal.
Type IV	**Mixed Type**	Exhibits Type II (pseudoarticulation) on one side and Type III (fusion) on the contralateral side.	**Clinical:** Creates asymmetric torsional forces on the disc, highly predisposing to herniation at the level above. **Labeling:** Requires checking both left and right sides on coronal views.

3.2 Visual Identification in Annotation Tools

When using a tool like Roboflow, the annotator must not rely solely on the Sagittal view (which is standard for counting). The Coronal view is the "truth" plane for Castellvi typing.

- **Search Pattern:** Scroll through the coronal stack. Identify the lowest lumbar-type vertebra. Examine the "wings" (transverse processes).
- **Measurement (Visual Approximation):** If the transverse process looks massive—often described as a "butterfly wing"—compare it to the L4 process above. If it is significantly broader (height >19mm), it is Type I at minimum.[10]
- **Articulation Check:** Zoom in on the interface between the transverse process and the sacrum.
  - *Is there a black line (cortical bone) separating them?* -> **Type II (Pseudoarticulation)**.
  - *Is the marrow signal continuous?* -> **Type III (Fusion)**.

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4. The O'Driscoll Classification: The Sagittal Standard

While Castellvi rules the coronal plane, many AI datasets consist of sagittal scouts or sagittal T2 sequences. In these cases, the O'Driscoll classification of the S1-S2 disc morphology is the primary tool for identifying S1 Lumbarization.[1]

4.1 The Importance of Disc Morphology

In a normal sacrum, the segments S1 through S5 are fused, with no disc material between them. However, in transitional anatomy, a "vestigial" or fully formed disc may appear between

S1 and S2.

- **Type 1:** No disc material. The S1 and S2 bodies are fused solid. (Normal Sacrum).
- **Type 2:** A small, rudimentary disc is present. Its anteroposterior (AP) length is significantly less than the AP length of the sacral body. This is a common variant but does not constitute full lumbarization.
- **Type 3:** A well-formed disc extends the **entire AP length** of the sacrum. This indicates that S1 is functionally separate from S2.
- **Type 4:** Similar to Type 3, but with **"Squaring" of the S1 body**.

4.2 The "Squaring" Sign

The shape of the vertebral body on the mid-sagittal MRI is a powerful predictor of its label.

- **Normal S1:** Wedge-shaped. The anterior height is greater than the posterior height, contributing to the sacral kyphosis.
- **Lumbarized S1 (L6):** Rectangular or "squared." The anterior and posterior heights are roughly equal. This morphology suggests the vertebra is acting as a load-bearing lumbar segment rather than part of the rigid sacrum.[1]

Annotation Protocol: If you encounter a vertebra that is the 6th counting down from L1, check its shape. If it is squared and sits above a full-length O'Driscoll Type 3/4 disc, it should be labeled as **L6** or **Transitional S1**, depending on the project's specific ontology.

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5. Protocols for Vertebral Counting and Landmarking

The central dilemma in spinal labeling is the reference point. Does one count down from the skull (C2) or up from the sacrum? The choice of method defines the accuracy of the labels.

5.1 The Gold Standard: Whole-Spine "Top-Down" Counting

Whenever whole-spine imaging (pan-spine MRI, stitched X-ray, or CT scouts) is available, the "Top-Down" method is mandatory. It is the only way to definitively identify the neurological levels that correspond to spinal nerve roots.[3]

Step-by-Step Top-Down Protocol:

1. **Identify C2 (The Axis):** Locate the Odontoid Process (Dens), the tooth-like projection extending superiorly into C1.
2. **Count Cervical (C1–C7):** Count down seven segments. C7 typically has a prominent spinous process.
3. **Count Thoracic (T1–T12):** T1 is defined as the first vertebra with articulating ribs.
    - *Crucial Check:* Does the patient have cervical ribs (at C7)? If so, do not mistake C7 for T1.

    - *Crucial Check:* Does the patient have 11 or 13 thoracic vertebrae? A Top-Down count respects the absolute number. If there are 13 rib-bearing segments, the 13th is T13.
4. **Count Lumbar (L1–L5):** The first non-rib-bearing vertebra is L1. Continue down.
    - If the count ends at 4 lumbar vertebrae before the sacrum -> **Sacralized L5**.
    - If the count ends at 6 lumbar vertebrae -> **Lumbarized S1**.

5.2 The Reality of Clinical Datasets: "Bottom-Up" Labeling

Most datasets used for training AI models (e.g., lumbar disc herniation detection) consist of **Isolated Lumbar Spine** scans. The cervical and upper thoracic spines are not visible. In this scenario, the annotator cannot count from C2. They must rely on probabilistic **Anatomical Landmarks** to infer the labels.

Warning: No single landmark is 100% accurate. A "Triangulation" approach using multiple landmarks is required to achieve >95% accuracy.[16]

5.2.1 Landmark 1: The Iliolumbar Ligament (ILL) – *The Anchor*

The Iliolumbar Ligament is widely regarded as the most reliable marker for identifying the L5 vertebra in isolated lumbar scans.

- **Anatomy:** The ILL connects the transverse process of L5 to the iliac crest. It functions to stabilize the lumbosacral junction.
- **Imaging Appearance:** Best seen on **Axial T1 or T2** MRI. It appears as a single or double hypointense (black) band extending laterally and posteriorly from the transverse process to the pelvis.
- **Statistical Reliability:**
    - In **93.8%** of normal cases, the ILL arises from **L5**.
    - In **Sacralization (L5 -> S1)**: The ILL often arises from **L4** (the last mobile vertebra).
    - In **Lumbarization (S1 -> L6)**: The ILL arises from **L6** (the transitional S1) or L5.[7]
- **Annotation Rule:** Find the vertebra with the ILL. Label it as L5 (or the lowest lumbar type). If this vertebra is fused to the sacrum, you have confirmed a Sacralized L5.

5.2.2 Landmark 2: The 12th Rib (Costal Facet) – *The Thoracic Limit*

Identifying the transition from Thoracic to Lumbar (T12 to L1) is critical for starting the lumbar count.

- **Imaging Appearance:** On **Sagittal T2** MRI, scroll to the most lateral slices (parasagittal). Look for the **Costovertebral Joint**: a small, dark, oval articulation on the posterolateral aspect of the vertebral body.
- **Statistical Reliability:**
    - In **96.9%** of normal subjects, the last rib is at T12.
    - **Pitfall (T11 Ribs):** In cases of L5 Sacralization, there is a higher incidence (30%) of having only 11 pairs of ribs. If you assume the last rib is T12, you will mislabel L1 as L2, shifting the whole count down.[16]

    - **Pitfall (Lumbar Ribs):** Small, rudimentary ribs can occur at L1 (Type 1 lumbar ribs). Differentiate them by seeing if they articulate (rib) or are fused (transverse process).
- **Annotation Rule:** Use the rib count as a *starting point*, but verify with the ILL at the bottom. If the count from the ribs conflicts with the count from the ILL (e.g., Rib count says L4, ILL says L5), **trust the ILL** or flag the case as "Indeterminate/Requires Whole Spine."

5.2.3 Landmark 3: Paraspinal Vascular Anatomy

The positions of major vessels are useful secondary confirmations but should not be used in isolation due to variability.[16]

- **Aortic Bifurcation:** Typically occurs at the level of the **L4** vertebral body.
    - *Shift:* In Sacralization, it often shifts cranially (L3-L4). In Lumbarization, it shifts caudally (L4-L5).
- **Right Renal Artery (RRA):** Typically originates at the **L1** vertebral body or L1-L2 disc.
- **Superior Mesenteric Artery (SMA):** Typically originates at T12 or T12-L1.
- **Psoas Muscle Origin:** On Coronal MRI, the psoas muscles typically originate at the **D12-L1** disc level, appearing as pointed tips.

Table 2: Reliability of Anatomical Landmarks for L5 Identification

[16]

Landmark	Typical Location	Reliability in Normal Spine	Behavior in Sacralization	Behavior in Lumbarization
Iliolumbar Ligament	L5	**High (~94%)**	Originates from **L4** (75-80% of cases)	Originates from **L6/S1** or L5
Last Costal Facet	T12	**High (~97%)**	Last rib often at **T11** (11 rib pairs)	Last rib at T12 (sometimes L1 lumbar rib)
Aortic Bifurcation	L4	Moderate (~50%)	Shifts Cranially (L3-L4)	Shifts Caudally (L4-L5)
Right Renal Artery	L1	Moderate (~55%)	Shifts Cranially (D12-L1)	Shifts Caudally (L2)

Psoas Origin	T12-L1	Moderate (~70%)	No significant shift	Caudal Shift (L1-L2)

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6. Imaging Modality Nuances: MRI vs. CT vs. X-Ray

Annotators must adapt their search patterns based on the physics of the image modality provided in the dataset.

6.1 Magnetic Resonance Imaging (MRI)

MRI is the dominant modality for spinal AI datasets (e.g., herniation detection).

- **T2-Weighted Sagittal:** The workhorse for labeling. Fluid (CSF) is bright white. Discs are bright (if healthy) or dark (if degenerated).
  - *LSTV Tip:* Look for the "Vestigial Disc" at S1-S2. On T2, a true disc will have a high-signal nucleus pulposus. A fused S1-S2 will be dark (marrow signal).[2]
- **T1-Weighted Axial:** Best for anatomy.
  - *LSTV Tip:* The **Iliolumbar Ligament** is best seen here as a dark band against the bright perirenal fat.
- **STIR (Short Tau Inversion Recovery):** Suppresses fat, making fluid/edema bright.
  - *LSTV Tip:* Critical for identifying **Bertolotti's Syndrome**. Active inflammation at a Type II pseudoarticulation will glow bright white on STIR sequences, indicating bone marrow edema.[10]

6.2 Computed Tomography (CT)

CT provides superior osseous detail but poor soft tissue contrast.

- **Bone Window:** Best for checking fusion.
  - *LSTV Tip:* A Type III fusion is confirmed if the trabecular bone pattern flows uninterrupted from the transverse process to the sacrum. On MRI, this can be ambiguous; on CT, it is definitive.
- **Coronal Reformats:** Essential for Castellvi typing. Annotators should always check if the dataset includes coronal reconstructions (MPR) to verify the transverse processes.

6.3 Plain Radiography (X-Ray)

- **Ferguson View:** An AP view angled 30 degrees cranially. This "opens up" the L5-S1 joint space.
  - *LSTV Tip:* This is the view for which the Castellvi classification was originally designed. The "butterfly" appearance of the transverse processes is most classic here.[1]

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7. Technical Annotation Manual: Workflow in Roboflow

This section provides a practical, step-by-step guide for using Roboflow to apply the anatomical knowledge detailed above. It addresses the specific tools (Bounding Box, Polygon, Keypoint) and workflow management required for high-throughput labeling.

7.1 Setting Up the Ontology (Class Management)

Before labeling, the dataset ontology must be defined to handle LSTV ambiguity.

- **Standard Classes:** vertebra_L1, vertebra_L2, vertebra_L3, vertebra_L4, vertebra_L5, vertebra_S1.
- **Transitional Classes:** vertebra_LSTV, or specific tags like LSTV_Type2, LSTV_Type3.
- **Landmark Classes:** iliolumbar_ligament, rib_T12.

Recommendation: Use "Attributes" in Roboflow. Label the bone as vertebra_L5 and add an attribute transitional: true or Castellvi: Type II. This allows the AI to learn the general shape of a vertebra while also learning the subtype features.

7.2 Bounding Box Annotation (Object Detection)

For training models like YOLOv8.[19]

1. **Tool Selection:** Press b to activate the Bounding Box tool.
2. **Boundary Definition:**
    - **Superior/Inferior:** Align the box edges exactly with the vertebral endplates. **Do not** include the intervertebral disc.
    - **Anterior/Posterior:** Align with the cortical margins of the vertebral body. **Do not** include osteophytes (bone spurs) if they extend significantly beyond the true corner, unless instructed otherwise.
    - **Posterior Elements:** Generally, exclude the pedicles and spinous processes unless the class is "Whole Vertebra." For "Vertebral Body" detection, the box should be tight to the corpus.
3. **Handling LSTV:**
    - If marking a **Sacralized L5**, draw the box around the body. If it is fused to S1, use the "waist" of the fusion mass (where the disc *should* be) as the inferior border.
    - If marking a **Lumbarized S1**, draw the box around the squared S1 body. Ensure the S1-S2 disc is excluded.

7.3 Polygon Annotation (Instance Segmentation)

For training models like Mask R-CNN or U-Net.[21]

1. **Tool Selection:** Press p for Polygon or s for Smart Polygon (SAM-based).
2. **Technique:**
    - Trace the cortical outline of the vertebral body.

    - **Smart Polygon Workflow:** Click inside the vertebra. The AI will generate a mask. Check the edges.
    - **Correction:** If the mask bleeds into the Psoas muscle (common on lateral views), hold Alt (Windows) or Option (Mac) and click the incorrect area to subtract it.
3. **LSTV Specifics:**
    - **Type II (Pseudoarticulation):** On Coronal views, you must precisely trace the gap between the transverse process and the sacrum. Do not create a single mask that bridges this gap. The gap is the defining feature of the pathology.
    - **Type III (Fusion):** If fully fused, the mask may need to encompass both the L5 transverse process and the sacral ala as a continuous unit, depending on the segmentation goal (bone separation vs. functional unit).

7.4 Keypoint Annotation (Pose Estimation)

For training models that measure spinal alignment (e.g., Cobb Angle, Lordosis).[23]

1. **Standard Protocol:** Place 4 points per vertebra (4-point method):
    - Anterior-Superior Corner.
    - Posterior-Superior Corner.
    - Anterior-Inferior Corner.
    - Posterior-Inferior Corner.
2. **LSTV Nuance:**
    - In a **Wedge-Shaped L5** (common in sacralization), the anterior height is much larger than the posterior. Place points on the *true* corners of the bone, even if the shape is distorted.
    - **Centroid:** Some protocols require a single center point. Place this at the geometric center of the vertebral body.

7.5 Collaborative Workflow & QA

LSTV labeling requires consensus.

1. **Batch Assignment:** Use Roboflow's "Assign" feature to distribute images. Assign difficult batches (e.g., "Potential LSTV") to senior annotators.[25]
2. **The "Null" Tool:** If an image is too blurry, artifact-heavy, or anatomically indeterminate (e.g., severe scoliosis preventing counting), press n to mark it as Null. Do not guess. Bad data is worse than no data.
3. **Comments:** Use the comment tool (c) to tag ambiguous cases. Example: "@Reviewer - T12 rib is missing, but ILL is at L4. Suspect 11 rib pairs + Sacralization. Please confirm.".[26]

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8. Common Pitfalls, Mimics, and How to Avoid Them

Medical students often confuse LSTV with acquired pathologies. Differentiating "Born with it"

(LSTV) from "Acquired it" (Disease) is vital.

8.1 Infection (Discitis/Osteomyelitis) vs. Type III Fusion

An acquired fusion from an old infection can look like a congenital Type III fusion.

- **Differentiation:**
  - **LSTV:** The fusion is smooth. The trabecular bone pattern is organized. The "waist" of the vertebra is preserved.
  - **Infection:** The fusion is chaotic. There is bone destruction, endplate erosion, and often a loss of normal vertebral height. The "disc space" looks eaten away, not congenitally absent.[27]

8.2 Surgical Fusion (Post-Instrumented) vs. Type III Fusion

A patient who had a previous L5-S1 fusion surgery (PLIF/TLIF) will have a fused joint.

- **Differentiation:** Look for susceptibility artifacts (black "blooming" holes) from screws or cages, even if the metal itself is not in the slice. Look for the laminectomy defect (missing spinous process) posteriorly. LSTV patients have intact posterior elements.

8.3 Compression Fracture vs. LSTV Wedging

The L5 vertebra in LSTV is often naturally wedge-shaped (anterior height > posterior height) to support the lordotic curve.

- **Differentiation:**
  - **Fracture:** Look for a "step-off" in the anterior cortex, a fracture line, or acute edema (bright on STIR). There is typically a *loss* of height compared to prior exams.
  - **LSTV:** The cortical margins are smooth and intact. The wedging is developmental and stable.[1]

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9. Quality Assurance and Inter-Rater Reliability

Inter-rater reliability (IRR) for LSTV detection is notoriously variable, with kappa values often ranging from 0.6 to 0.8. To maximize IRR in your dataset:

1. **Consensus Protocol:** Implement a "Double-Blind" labeling strategy for the first 100 cases. Two students label the same images independently. A senior radiologist reviews discrepancies.
2. **Gold Standard Set:** Create a "Reference Set" of 50 images with confirmed LSTV types (verified by 3D CT or whole-spine imaging). New annotators must label this set and achieve >90% IoU (Intersection over Union) before labeling the main dataset.
3. **Review Loop:** The "Review" stage in Roboflow is critical. The reviewer should check the **Class Distribution**. If the dataset has 0% LSTV, the annotators are missing them (since

prevalence is ~15%). If it has 50% LSTV, they are over-calling normal variants.

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10. AI Implications: Why This Matters

The labels you create are the "ground truth" for AI.

- **Object Detection (YOLO):** If you label a Sacralized L5 as "S1" in one image and "L5" in another, the model learns that "L5" has highly variable features. This increases the model's loss function and reduces detection confidence. Consistent labeling (e.g., always labeling the functional L5 as L5, regardless of fusion) helps the model converge.[5]
- **Segmentation (U-Net):** Including the ILL or osteophytes in your polygon mask will cause the model to over-segment, potentially leading to errors in automated Cobb angle measurements or stenosis grading.[5]

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11. Conclusion

The accurate labeling of the vertebrae, particularly in the presence of Lumbosacral Transitional Vertebrae, is a task that requires more than simple counting. It demands a sophisticated understanding of embryology, biomechanics, and radiological physics. By adopting the multi-landmark "Triangulation" approach—verifying the T12 rib, anchoring with the Iliolumbar Ligament, and confirming with Sagittal/Coronal morphology—medical student annotators can produce datasets of clinical grade.

Final Checklist for the Annotator:

1. **Scan:** Check the scout/localizer. Can you count from C2?
2. **Anchor:** Find the Iliolumbar Ligament on Axial view. This is your provisional L5.
3. **Verify:** Check the sagittal view for S1 squaring and the T12 rib.
4. **Classify:** Check the coronal view for Castellvi features (butterfly wings).
5. **Label:** Apply the bounding box/polygon strictly according to the defined ontology, using attributes for transitional states.

Adherence to this manual ensures that the digital twins of human anatomy used to train the next generation of AI tools are as accurate, safe, and reliable as the medicine they are intended to support.

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